Quantum statistical mechanics : 2-particle example

Probability of finding two identical particles $\sim |\psi(x_1, x_2)|^2 = |\psi(x_2, x_1)|$

$\Rightarrow$ (for a single-valued function) $\psi(x_1, x_2) = \pm \psi(x_2, x_1)$

$\Rightarrow$ <u>Fermions</u> : antisymmetric w.f.

$$\psi_F(x_1, x_2) = -\psi_F(x_2, x_1)$$

$\Rightarrow$ <u>Bosons</u> : symmetric w.f.

$$\psi_B(x_1, x_2) = \psi_B(x_2, x_1)$$

consider <u>Free particles</u> : $\hat{H} = \sum_{j=1}^{N} \left(-\frac{\hbar^2}{2m} \vec{\nabla}_j^2 \right)$

clearly $\hat{H}$ is symmetric under any permutation of particles.

1) Eigenfunction of $\hat{H}_1 = -\frac{\hbar^2}{2m} \vec{\nabla}_1^2$ is $\psi(x_1) = \langle \vec{x}_1 | \vec{k} \rangle =$ (plane wave) $= \frac{1}{\sqrt{V}} e^{i \vec{k} \vec{x}_1}$

$$\hat{H}_1 \frac{1}{\sqrt{V}} e^{i \vec{k} \vec{x}_1} = \frac{\hbar^2 k^2}{2m} \frac{1}{\sqrt{V}} e^{i \vec{k} \vec{x}_1}$$

$$\varepsilon(k_1) = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m} \quad \text{(K.E. of free particle)}$$

2) Two particles

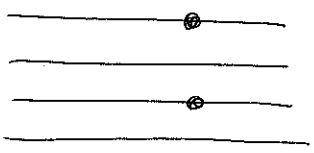
<u>Fermions</u> (spinless) $\psi_F(x_1, x_2) \rightarrow \frac{1}{\sqrt{2}} \left(|\vec{k}_1\rangle |\vec{k}_2\rangle - |\vec{k}_2\rangle |\vec{k}_1\rangle \right)$

$$= \frac{1}{\sqrt{2}} \frac{1}{V} \left(e^{i k_1 x_1 + i k_2 x_2} - e^{i k_1 x_2 + i k_2 x_1} \right)$$

Observe that $\psi_F(x_1, x_1) = 0 \leftarrow$ (Pauli principle)

Also $|k_1\rangle |k_1\rangle - |k_1\rangle |k_1\rangle = 0$

$$\hat{H}_{1+2}\,\psi_F(x_1,x_2) = \frac{\hbar^2}{2m}\left(k_1^2+k_2^2\right)\psi_F(x_1,x_2)$$



every state is uniquely specified by <u>occupied</u> levels ε_1 and ε_2 (equivalently, by occupied momenta $\vec{k}_1$ and $\vec{k}_2$).

Partition function for two <u>spinless electrons</u>:

$$Z_F = \sum_{\substack{\text{antisymmetric}\\ \text{states}}} e^{-\varepsilon/\tau} =$$

$$= \sum_{k_1,k_2} \langle k_1,k_2|\, e^{-\frac{\hat{H}}{\tau}}\, |k_1,k_2\rangle =$$

$$= \sum_{k_1<k_2} \frac{1}{\sqrt{2}}\left(\langle k_1|\langle k_2| - \langle k_2|\langle k_1|\right) e^{-\frac{\hat{H}}{\tau}} \frac{1}{\sqrt{2}}\left(|k_1\rangle|k_2\rangle - |k_2\rangle|k_1\rangle\right)$$

$$= \frac{1}{2}\sum_{k_1<k_2}\left\{\left(\langle k_1|\langle k_2|\right)\left(|k_1\rangle|k_2\rangle\right) + \left(\langle k_2|\langle k_1|\right)\left(|k_2\rangle|k_1\rangle\right)\right\} \cdot$$

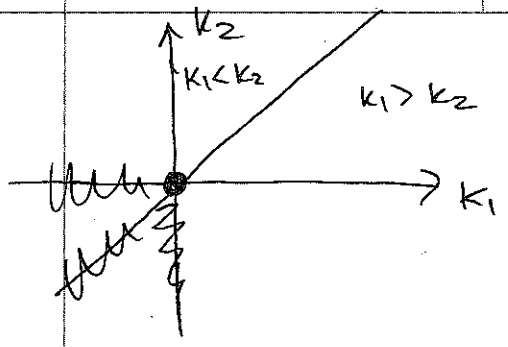
$$\cdot \exp\left[-\frac{\hbar^2}{2m\tau}\left(k_1^2+k_2^2\right)\right]$$

need this ordering in order not to overcount!

$$= \sum_{k_1<k_2} \exp\left[-\frac{\hbar^2}{2m\tau}\left(k_1^2+k_2^2\right)\right].$$

Observe that states $\left\{\begin{matrix}|5\rangle|7\rangle - |7\rangle|5\rangle \\ |7\rangle|5\rangle - |5\rangle|7\rangle\end{matrix}\right\}$ are in fact one state since $\left(|5\rangle|7\rangle - |7\rangle|5\rangle\right) = -\left(|7\rangle|5\rangle - |5\rangle|7\rangle\right)$

$\Rightarrow$ in order not to overcount momenta $\vec{k}_1, \vec{k}_2$ have to be ordered: for example, $|k_1| < |k_2|$.



The restriction $(k_1 < k_2)$ can be incorporated in a more convenient way:

$$Z_F = \frac{1}{2}\left\{ \sum_{k_1, k_2} e^{-\frac{\hbar^2}{2m\tau}(k_1^2+k_2^2)} - \sum_{k_1=k_2} e^{-\frac{\hbar^2}{2m\tau}(k_1^2+k_1^2)} \right\}$$

no restriction here

$$= \frac{1}{2}\left\{ \left(\sum_{k_1} e^{-\frac{\hbar^2 k_1^2}{2m\tau}}\right)\left(\sum_{k_2} e^{-\frac{\hbar^2 k_2^2}{2m\tau}}\right) - \left(\sum_{k} e^{-\frac{\hbar^2 k^2}{m\tau}}\right)\right\}$$

But $Z_1 = \sum_{k_1} e^{-\frac{\hbar^2 k_1^2}{2m\tau}} = \frac{V}{(2\pi)^3}\int d^3\vec{k_1}\, e^{-\frac{\hbar^2 k_1^2}{2m\tau}} =$

$$= \frac{V}{(2\pi)^3}\, 4\pi \int_0^\infty dk_1\, k_1^2\, e^{-\frac{\hbar^2 k_1^2}{2m\tau}} =$$

$$k_1 = \sqrt{\frac{2m\tau}{\hbar^2}}\, x$$

$$= \frac{V}{8\pi^3}\, 4\pi \left(\frac{2m\tau}{\hbar^2}\right)^{3/2} \underbrace{\int_0^\infty dx\, x^2 e^{-x^2}}_{\frac{\sqrt{\pi}}{4}} =$$

$$= \frac{V}{8\pi^{3/2}}\left(\frac{2m\tau}{\hbar^2}\right)^{3/2} = V\left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2} = V\left(\frac{2\pi m\tau}{h^2}\right)^{3/2}$$

$$\Rightarrow Z_1 = \frac{V}{\lambda^3} \quad \text{where thermal wavelength}$$

$$\lambda = \frac{h}{\sqrt{2\pi m\tau}} = \frac{2\pi\hbar}{\sqrt{2\pi m\tau}}$$

(just as before)

$$\Rightarrow Z_F = \frac{1}{2}\left\{Z_1^2(m) - Z_1\left(\frac{m}{2}\right)\right\}$$

$$Z_1(m) = V\left(\frac{2\pi m\tau}{h^2}\right)^{3/2}, \quad Z_1\left(\frac{m}{2}\right) = V\left(\frac{2\pi\left(\frac{m}{2}\right)\tau}{h^2}\right)^{3/2}.$$

<u>Bosons</u> $\quad \psi_B(x_1, x_2) \to |k_1, k_2\rangle_B = \begin{cases} \frac{1}{\sqrt{2}}\, |k_1\rangle|k_2\rangle + |k_2\rangle|k_1\rangle & (k_1 < k_2) \\ |k_1\rangle|k_2\rangle & (k_1 = k_2) \end{cases}$

$$\Rightarrow Z_B = \sum_{\substack{\text{symmetric}\\ \text{states}}} e^{-\varepsilon/\tau} =$$

$$= \sum_{k_1, k_2} {}_B\langle k_1, k_2|\, e^{-\hat{H}/\tau}\, |k_1, k_2\rangle_B =$$

$$= \sum_{k_1 < k_2} \frac{1}{\sqrt{2}}\left(|k_1\rangle|k_2\rangle + |k_2\rangle|k_1\rangle\right) e^{-\hat{H}/\tau} \frac{1}{\sqrt{2}}\left(|k_1\rangle|k_2\rangle + |k_2\rangle|k_1\rangle\right)$$

$$+ \sum_{k_1 = k_2} \langle k_1|\langle k_2|\, e^{-\hat{H}/\tau}\, |k_1\rangle|k_2\rangle =$$

$$= \sum_{k_1 < k_2} e^{-\frac{\hbar^2}{2m\tau}(k_1^2 + k_2^2)} + \sum_{k_1} e^{-\frac{\hbar^2}{2m\tau}(2k_1^2)} =$$

$$= \frac{1}{2}\left\{\sum_{k_1, k_2} e^{-\frac{\hbar^2(k_1^2 + k_2^2)}{2m\tau}} - \sum_{k_1 = k_2} e^{-\frac{\hbar^2 2k_1^2}{2m\tau}}\right\} + \sum_{k_1} e^{-\frac{\hbar^2 k_1^2}{m\tau}} =$$

$$Z_B = \frac{1}{2}\left\{\left(\sum_{k_1} e^{-\frac{\hbar^2 k_1^2}{2m\tau}}\right)\left(\sum_{k_2} e^{-\frac{\hbar^2 k_2^2}{2m\tau}}\right) + \sum_{k} e^{-\frac{\hbar^2 k^2}{m\tau}}\right\} =$$

$$= \frac{1}{2}\left\{Z_1^2(m) + Z_1\left(\frac{m}{2}\right)\right\}.$$

Thus
$$\begin{cases} Z_F = \frac{1}{2}\left\{\left(Z_1(m)\right)^2 - Z_1\left(\frac{m}{2}\right)\right\} \\ Z_B = \frac{1}{2}\left\{\left(Z_1(m)\right)^2 + Z_1\left(\frac{m}{2}\right)\right\} \end{cases}$$

while two <u>classical</u> identical particles are described by

$$Z_N = \frac{(Z_1)^N}{N!} \longrightarrow Z_2 = \frac{(Z_1)^2}{2!} = \frac{1}{2}\left(Z_1(m)\right)^2.$$

$\Rightarrow$ Results for <u>quantum</u> particles generally differs from <u>classical</u>!

$\ast$ Observe that $Z_{F,B} = \frac{1}{2}\left(Z_1(m)\right)^2 \mp \frac{1}{2} Z_1\left(\frac{m}{2}\right)$ can be approximated by classical $Z_2 = \frac{1}{2}\left(Z_1(m)\right)^2$ when the quantum correction $\left[\mp \frac{1}{2} Z_1\left(\frac{m}{2}\right)\right]$ is small in comparison with the first (Z_2) term:

that is, if $Z_1\left(\frac{m}{2}\right) \ll \left(Z_1(m)\right)^2$

$$V\left(\frac{\pi m\tau}{h^2}\right)^{3/2} \ll \left[V\left(\frac{2\pi m\tau}{h^2}\right)^{3/2}\right]^2 =$$

$$= V^2\left(\frac{2\pi m\tau}{h^2}\right)^3$$

Thus $\left(\frac{h^2}{2\pi m \tau}\right)^{3/2} \ll V$

$$\lambda^3 \ll V$$

Quantum effects small when thermal length is much smaller than the size of the box.

But since $\lambda = \frac{h}{\sqrt{2\pi m \tau}}$ is function of temperature,

at sufficiently low τ unavoidably $\lambda^3 \sim V$ and then classical approximation breaks!

That happens when $\lambda^3 \approx V$ $\quad (V = L^3)$

$$\frac{h^2}{2\pi m \tau} \approx V^{2/3} = L^2$$

$$\tau \approx \frac{h^2}{2\pi m L^2}$$ (~ level spacing in the box of size L.)

Grand partition function:

quantum state is uniquely determined by occupation numbers $\{n_k\}$ of different states in k-space.

For fermions: $n_k = 0, 1$.

For bosons: $n_k = 0, 1, 2, \ldots \infty$

Then for ideal gas of particles: $\varepsilon_s \to \varepsilon(\vec{k})\, n_k$

meaning that each of n_k particles in state $\vec{k}$ carries energy $\varepsilon(\vec{k})$.

grand partition

$$Z_G = \sum_{N=0}^{\infty} e^{\frac{\mu N}{\tau}} \sum_{\{n_k\}} \prod_k e^{-\frac{\varepsilon(k)\, n_k}{\tau}}$$

$\hookrightarrow$ different k's contribute independently.

here $\sum_k n_k = N$ but the sum over $\sum_N$ ~~allows us to~~ removes this restriction:

$$Z_G = \sum_{n_k} \prod_k \exp\left(-\frac{\varepsilon_k n_k - \mu n_k}{\tau}\right) =$$

$$= \prod_k \sum_{n_k} \exp\left[-\frac{(\varepsilon_k - \mu)}{\tau} n_k\right]$$

- Fermions: $Z_G^F = \prod_k \left(1 + e^{\frac{\mu - \varepsilon_k}{\tau}}\right)$

$$\sum_{n_k=0}^{1}$$

- Bosons:

$$\sum_{n_k=0}^{\infty} \left(e^{\frac{\mu-\varepsilon_k}{\tau}}\right)^{n_k} = \frac{1}{1 - e^{\frac{\mu-\varepsilon_k}{\tau}}}$$

$$\Rightarrow Z_G^B = \prod_k \left(1 - e^{\frac{\mu-\varepsilon_k}{\tau}}\right)^{-1}$$

$$\Rightarrow \ln Z_G^{F/B} = -\eta \sum_k \ln\left(1 - \eta \exp\left[\frac{\mu-\varepsilon_k}{\tau}\right]\right) \qquad \eta = \begin{cases} -1 & \text{fermions} \\ +1 & \text{bosons} \end{cases}$$

Average occupation number of a state of energy $\varepsilon(k)$ is

- Fermions : $\langle n_k \rangle = \prod_{k'} \sum_{\{n_{k'}\}} n_k \frac{e^{\frac{\mu-\varepsilon_{k'}}{\tau} n_{k'}}}{Z_G^F} =$ (only $k'=k$ matters)

$$= \frac{0 + 1 \cdot e^{\frac{\mu-\varepsilon_k}{\tau}}}{1+e^{\frac{\mu-\varepsilon_k}{\tau}}} = \frac{1}{e^{\frac{\varepsilon_k-\mu}{\tau}}+1}$$ <u>Fermi-Dirac</u>

- Bosons : $\langle n_k \rangle = \sum_{n_k} \frac{n_k \left(e^{\frac{\mu-\varepsilon_k}{\tau}}\right)^{n_k}}{Z_G^B} = \left(\text{let } \frac{\mu-\varepsilon_k}{\tau} = x\right) =$

$$= \frac{\sum_n n e^{xn}}{\sum e^{xn}} = \frac{\frac{\partial}{\partial x}\sum_n e^{xn}}{\frac{1}{1-e^x}} = (1-e^x)\frac{\partial}{\partial x}\frac{1}{1-e^x} =$$

$$= (1-e^x)\frac{e^x}{(1-e^x)^2} = \frac{e^x}{1-e^x} = \frac{1}{e^{-x}-1} = \frac{1}{e^{\frac{\varepsilon-\mu}{\tau}}-1}$$

$$\Rightarrow \langle n_k \rangle = \frac{1}{e^{\frac{\varepsilon_k-\mu}{\tau}}-1}$$ Bose-Einstein